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Section - A

Meqs :-

- | | |
|---|---|
| ① | C |
| ② | C |
| ③ | B |
| ④ | B |
| ⑤ | B |
| ⑥ | B |
| ⑦ | C |
| ⑧ | B |
| ⑨ | D |
| ⑩ | C |
| ⑪ | C |
| ⑫ | D |

Section - B

Q.2(i)(or)

Ans.2(i)(or)

Data :-

force : 150N

displacement : 5m

workdone :- ?

$150\text{N} \times 5\text{m}$

$750\text{Nm} / 750\text{J}$

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Q.2 (ii) (or) _____
A.2 (ii) (or) _____

Data:

mass = 120g $\rightarrow$ Kg (120 $\div$ 1000 = 0.12Kg) $K_E = 50J$ Speed: $v = ?$

$$K_E = \frac{1}{2} m v^2$$

$$50J = \frac{1}{2} (0.12)(v)^2$$

$$\frac{50 \times 2}{0.12} = v^2$$

$$\frac{100}{0.12} = v^2$$

$$833 = v^2$$

$$\sqrt{833} = \sqrt{v^2}$$

$$28.7 = v$$

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Q: 2(iii)(or)

A: 2(iii)(or)

mass :- 60 kg

height :- 1.5 m

 $g = 9.8$ $P_E = mgh$

$$60 \times 9.8 \times 1.5$$
$$= 882 \text{ J}$$

Q: 2(iv)(or)

A: 2(iv)(or)

mass :- 0.45 kg

speed :- 22 m/s

 $g = 9.8$ $h = ?$ $K_E = P_E$

$$\frac{1}{2}mv^2 = mgh$$

$$\frac{1}{2}v^2 = gh$$

$$\frac{v^2}{g} = h \Rightarrow \frac{(22)^2}{9.8 \times 2} = h$$

$$\Rightarrow \frac{484}{19.6} = h \Rightarrow h = 24.7 \text{ m}$$

Q: 2(v)

A: 2(v)

(a) Generation of electricity:-

Potential energy of water is stored at the certain height ~~of~~ is converted into kinetic energy making it fall on the turbine to produce electricity.

(b) Cyclist going up to the hill:-

Stored chemical energy on the body of the cyclist allows him to do work against gravity. At the top of hill when he possesses the gravitational potential energy which allows him to go down the hill with increasing kinetic energy and even without pedaling.

Q: 2(vi)

A: 2(vi)

Renewable resources	Non renewable resources
① Renewable resources are the resources that can be replaced by a natural process.	Non-renewable resources are the resources resources

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For Example:

- ⇒ Sunlight
- ⇒ Flowing water

that can are limited and cannot be replaced naturally in a short time.

For Example:

- ⇒ Coal
- ⇒ oil or natural gas.

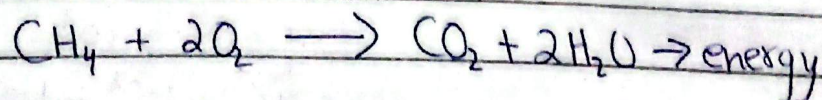
Q: 2(vii)

Ans: 2(vii)

fossil fuels :-

fossil fuels are the remain one million years old plant life (coal) and aquatic animal life (now gasoline and oil). include coal, oil and natural gas are rich in the hydrocarbon - molecules are made up of hydrogen and carbon.

Equation :-



Three types of fossil fuels :-

- ① Coal
- ② Natural gas
- ③ Oil

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Q: 2 (viii) (or)
 A: 2 (viii) (or)

Refination :-

Energy that directly convert sunlight into electricity is known as solar radiation.

Two ways for harnessing :-

- ① Solar photovoltaic cells.
- ② Thermal solar system.

Q: 2 (ix) (or)

A: 2 (ix) (or)

$$W = 3.69 \times 10^7$$

$$t = 1 \text{ m to sec}$$

$$1 \text{ m} = 60 \text{ sec}$$

$$P = \frac{W}{t} \Rightarrow \frac{3.69 \times 10^7}{60 \text{ sec}}$$

$$= \frac{0.0615 \times 10^7}{6.15 \times 10^{-2} \times 10^7}$$

$$\Rightarrow 6.15 \times 10^5$$

$$\text{hp} :- \frac{6.15 \times 10^5}{746}$$

$$= 825 \text{ hp}$$

Q: 2(n)(or)
A: 2(n)(or)

Data :-

$$E_{in} = 2238 \text{ J}$$

$$m = 10 \text{ kg}$$

$$h = 10 \text{ m}$$

$$g = 9.8$$

Output energy :-

$$\begin{aligned} mgh &= 10 \times 9.8 \times 10 \\ &= 980 \text{ J} \end{aligned}$$

Efficiency :-

$$\eta = \frac{E_{out}}{E_{in}} \times 100\%$$

$$\frac{980}{2238} \times 100$$

$$\eta = 43.8\%$$

Q: 2(ni)(or)
A: 2(ni)(or)

Data :-

$$m = 5 \text{ kg} \quad , \quad h = 12 \text{ m} \quad , \quad g = 9.8$$

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$$E_p = mgh$$
$$= 5 \times 9.8 \times 12$$
$$= 588 \text{ J}$$

speed =

$$K_e = \frac{1}{2} mv^2$$

$$588 = \frac{1}{2} (5) (v^2)$$

$$\frac{588 \times 2}{5} = v^2$$

$$\frac{1176}{5} = v^2$$

$$\frac{235.2}{\sqrt{235.2}} = \frac{v^2}{\sqrt{v^2}}$$

$$15.33 \text{ m/s} = v$$

①

②

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Section - C

Q: 3(i)

Ans: 3(i)

Defination :-

Work is said to be done when a force displaces a body on it's own direction.

Equation :-

$$\text{Workdone} = Fd$$

(force $\times$ displacement).

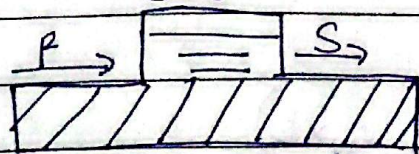
Work is a scalar quantity and it's SI unit is Joules ($1J = 1N \cdot m$).

Two conditions for zero work :-

① Parallel (Force and displacement) :-

force and displacement are parallel to each other.

$$\theta = 0^\circ$$



$$W = FS \cos \theta$$

$$= FS \cos(0^\circ)$$

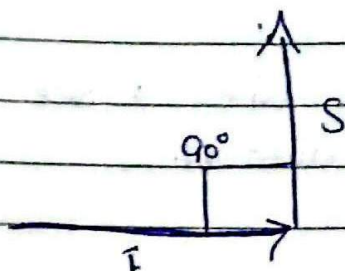
$$= FS(1)$$

$$\Rightarrow FS$$

(2) Perpendicular :-

when force and displacement are perpendicular to each other.

$$\theta = 90^\circ$$



$$\begin{aligned} W &= FS \cos \theta \\ &= FS \cos (90^\circ) \\ &= FS (0) \\ &= 0 \end{aligned}$$

Derivation of kinetic energy :-

$$K_E = \frac{1}{2} mv^2 \quad \left. \vphantom{\frac{1}{2} mv^2} \right\} s = v \times t$$

$$E_k = \text{work done}$$

$$\text{work done} = Fd$$

$$\frac{1}{2} mv^2 = (ma) \times (v \times t)$$

$$\frac{1}{2} mv^2 = m \left(\frac{v}{t} \right) \times \left(\frac{v}{2} \right) t$$

$$\frac{1}{2} mv^2 = m \left(\frac{v^2}{2} \right)$$

$$\frac{1}{2}mv^2 = \frac{1}{2}mv^2$$

④

Work - Energy theorem:

The work done on an object equals the change in its kinetic energy :-

$$W = \Delta E_K$$

Numerical :-

Data :-

mass :- 2.0 kg

height :- (total) 20m
(fallen) 15m

height fallen so far :- 15m

remaining :- 20m - 15m
= 5m.

①

Kinetic energy :-

$$E_K = P_E$$

$$E_K = mgh_{\text{fallen}}$$

$$= 2 \times 9.8 \times 15$$

$$= 294 \text{ J}$$

remaining PE :- $mgh_{\text{remaining}}$
 $\Rightarrow 2 \times 9.8 \times 5 = 98 \text{ J}$

$$\text{Total} = 294 + 98 = 392 \text{ J}$$

$$mgh_{\text{total}} = 2 \times 9.8 \times 20 \text{ m} = 392 \text{ J} \quad \checkmark_{\text{conserved}}$$

$$\begin{array}{c} \text{Q.3(ii)(or)} \\ \text{A.3(ii)(or)} \end{array}$$

Definition :-

Energy conversion refers to changing one form to another.
for example :-

① Generation of electricity :-

Potential energy of water is stored at the certain height is converted into kinetic energy making it fall on the turbine to produce the electricity.

② Cyclist going up to the hill :-

Stored chemical energy in a body of the cyclist allowing him to do work against gravity. At the top of the hill, he possesses the gravitational potential energy which allows him to go downhill with increasing kinetic energy without even pedaling.

Only 17-21% of chemical energy moves the car. The remaining 80% is lost as :-

heat from the engine, exhaust gas, and friction in mechanical parts. Scientists call this "dissipated energy".

Q.3(iii)

Ans.3(iii)

Power :-

Power is the time rate at which work is done or energy is consumed.

$$P = \frac{W}{t} \quad \text{or} \quad \frac{E}{t}$$

SI unit :- Watt (W), introduced after James watt. Scottish physicist who invented the first steam engine.

$$1W = 1Js^{-1}$$

Kilowatt - hour (kWh) :-

An unit of energy used in electricity billing. It is the energy converted ~~for~~ or ~~consumed~~ at the ^{constant} rate of 1 kW for 1 hour.

③ R Numerical.

Given:

~~mass: 2~~

P: ~~1960~~ 1960 W

h: 2m

t: 3s

g: 9.8 m/s

workdone :- $P \times t = \text{J}$

$$1960 \times 3s$$

$$= 5880 \text{ J}$$

$$W = mgh$$

$$\times 9.8$$

~~$$m = \frac{W}{gh}$$~~

$$m = \frac{W}{gh}$$

$$gh$$

$$= \frac{5880}{9.8 \times 2}$$

$$= \frac{5880}{19.6}$$

$$= 300 \text{ kg}$$

$$19.6$$

$$M = 300 \text{ kg}$$

Convert 1960W into hp

$$P = 1960 \times \frac{1}{746}$$

$$746$$

$$= \frac{1960}{746}$$

$$= 2.62 \text{ hp}$$

Q.3(iv)

A.3(iv)

Renewable
resourcesNon-renewable
resources.

Definition

i) Renewable resources

is the resource that
~~cannot~~ be replaced naturally
in a short time

ii) Non-renewable

resources that are
limited and cannot
be replaced rapidly
by a natural process.

Examples

① Sunlight

② Solar radiation

③ Wind , ④ flowing water

① Coal

② fossil fuels

③ Oil , ④ Natural gas.

fossil fuels :-

fossil fuels are the million
years old plant life (now coal) and
aquatic life now (gasoline), oil, natural gas).
Coal, oil and natural gas are rich in
hydrocarbons - molecules made up of
hydrogen and carbon.

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There are three types of fossil fuels:-

- ① Coal
- ② Oil
- ③ Natural gas.

Coal is mined.

Equation of fossil fuels:-

